

Task 2: Session 2 (AIML) - Assignment Questions

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1. if Statement

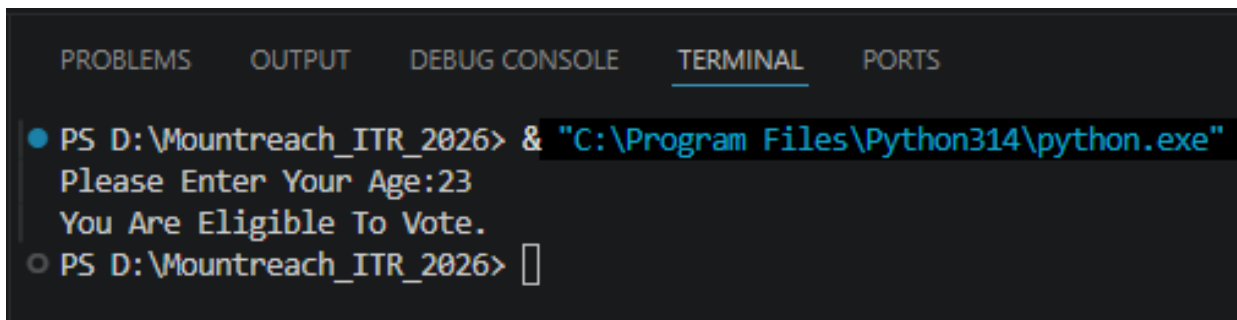
Write a program that asks the user to enter their age.

If the age is **18 or more**, print the message: "You are eligible to vote."

(Do not print anything if the condition is False).

Add comments.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Please Enter Your Age:23
You Are Eligible To Vote.
PS D:\Mountreach_ITR_2026> 
```

Program Code:

```
# Take Age As Input From User & Convert It To Integer
age = int(input("Please Enter Your Age:"))

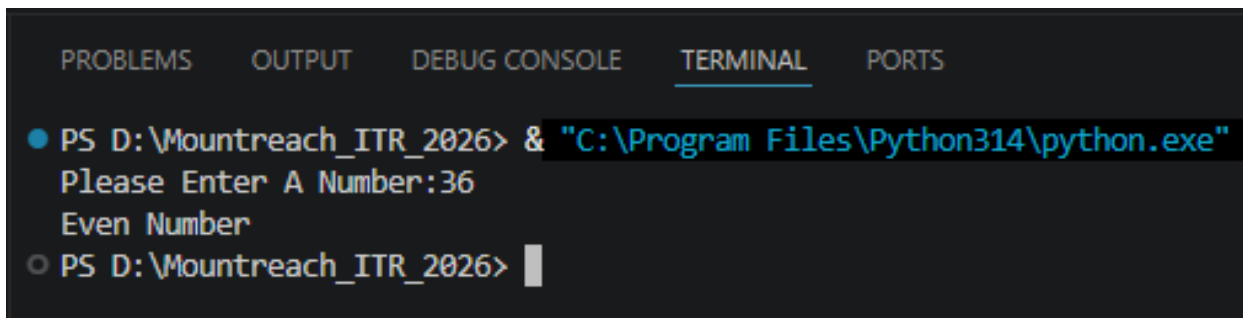
# Check If The Age Is 18 Or More
if age >= 18:
    print("You Are Eligible To Vote.") # Print Message Only If The Condition Is True
```

2. if - else Statement

Write a program that takes a number as input from the user and checks whether it is **even** or **odd**.

Print "Even Number" or "Odd Number" accordingly.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Please Enter A Number:36
Even Number
○ PS D:\Mountreach_ITR_2026> |
```

Program Code:

```
# Take Number As Input From User & Convert It To Integer
number = int(input("Please Enter A Number:"))

# Check If The Number Is Divisible By 2
if number % 2 == 0:
    print("Even Number") # If Remainder Is 0, So The Number Is Even
else:
    print("Odd Number") # Or ELSE The Number Is Odd
```

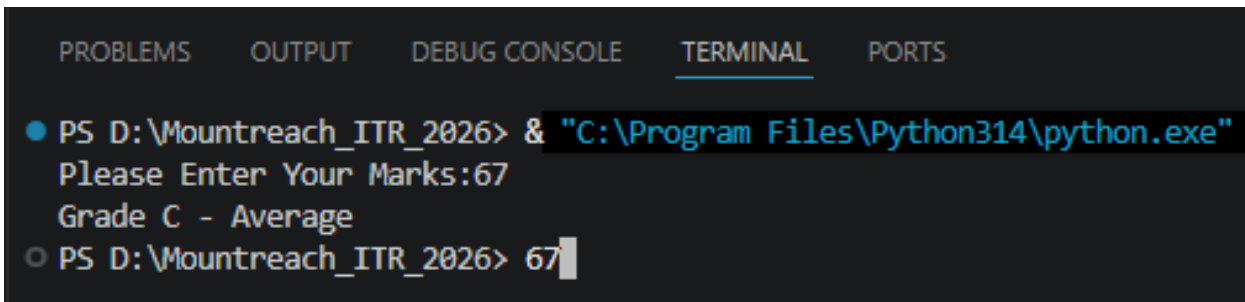
3. if - elif - else Statement

Write a program to assign grades based on marks entered by the user:

- Marks $\geq 90 \rightarrow$ "Grade A - Excellent"
- Marks $\geq 75 \rightarrow$ "Grade B - Good"
- Marks $\geq 60 \rightarrow$ "Grade C - Average"
- Marks $\geq 40 \rightarrow$ "Grade D - Pass"
- Marks $< 40 \rightarrow$ "Fail"

Use if-elif-else and print the appropriate grade message.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Please Enter Your Marks:67
Grade C - Average
○ PS D:\Mountreach_ITR_2026> 67
```

Program Code:

```
# Take Marks As Input From User & Convert It To Integer
marks = int(input("Please Enter Your Marks:"))

# Assigning Grade Based On Marks
if marks >= 90:
    print("Grade A - Excellent") # If Marks Are 90 Or Above -> Grade A
elif marks >= 75:
    print("Grade B - Good")      # If Marks Are Between 75 & 89 -> Grade B
elif marks >= 60:
    print("Grade C - Average")   # If Marks Are Between 60 & 74 -> Grade C
elif marks >= 40:
    print("Grade D - Pass")      # If Marks Are Between 40 & 59 -> Grade D
else:
    print("Fail")               # Or Else Marks Are Below 40 -> Fail
```

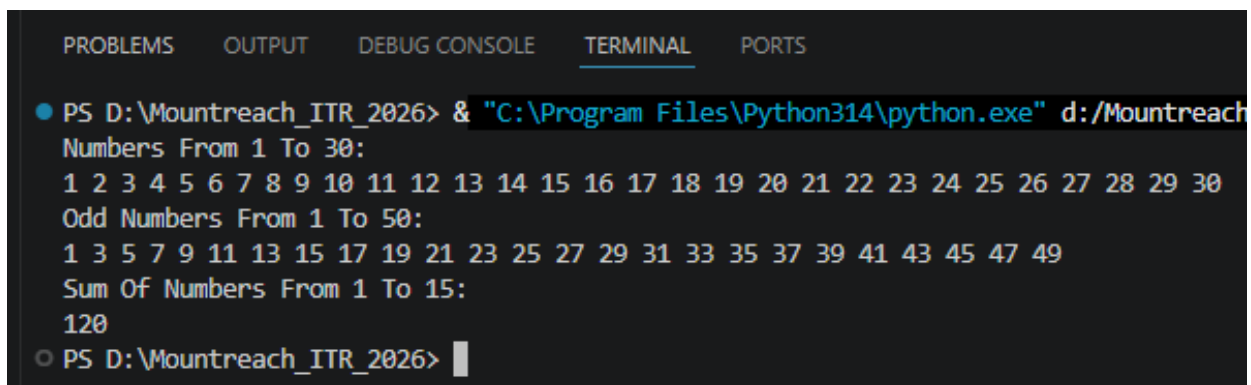
4. for Loop With Range

Write a program using for loop and range() to:

- Print all numbers from 1 to 30.
- Print all odd numbers from 1 to 50.
- Print the sum of numbers from 1 to 15.

Add comments explaining the use of range().

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe" d:/Mountreach
Numbers From 1 To 30:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
Odd Numbers From 1 To 50:
1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49
Sum Of Numbers From 1 To 15:
120
PS D:\Mountreach_ITR_2026> |
```

Program Code:

```
# Print All Numbers From 1 To 30
print("Numbers From 1 To 30:")
for number in range(1, 31):    # Starts At 1, Stops Before 31
    print(number, end=" ")

# Print All Odd Numbers From 1 To 50
print("\nOdd Numbers From 1 To 50:")
for number in range(1, 51, 2): # Starts At 1, Stops Before 51, Step Size 2
    print(number, end=" ")
(Only Odd Numbers)

# Print The Sum Of Numbers From 1 To 15
print("\nSum Of Numbers From 1 To 15:")
sum = 0
for number in range(1, 16):    # Starts At 1, Stops Before 16
    sum = sum + number         # Add Each Number To The Running Total
print(sum)                    # Print The Final Sum After The Loop Ends
```

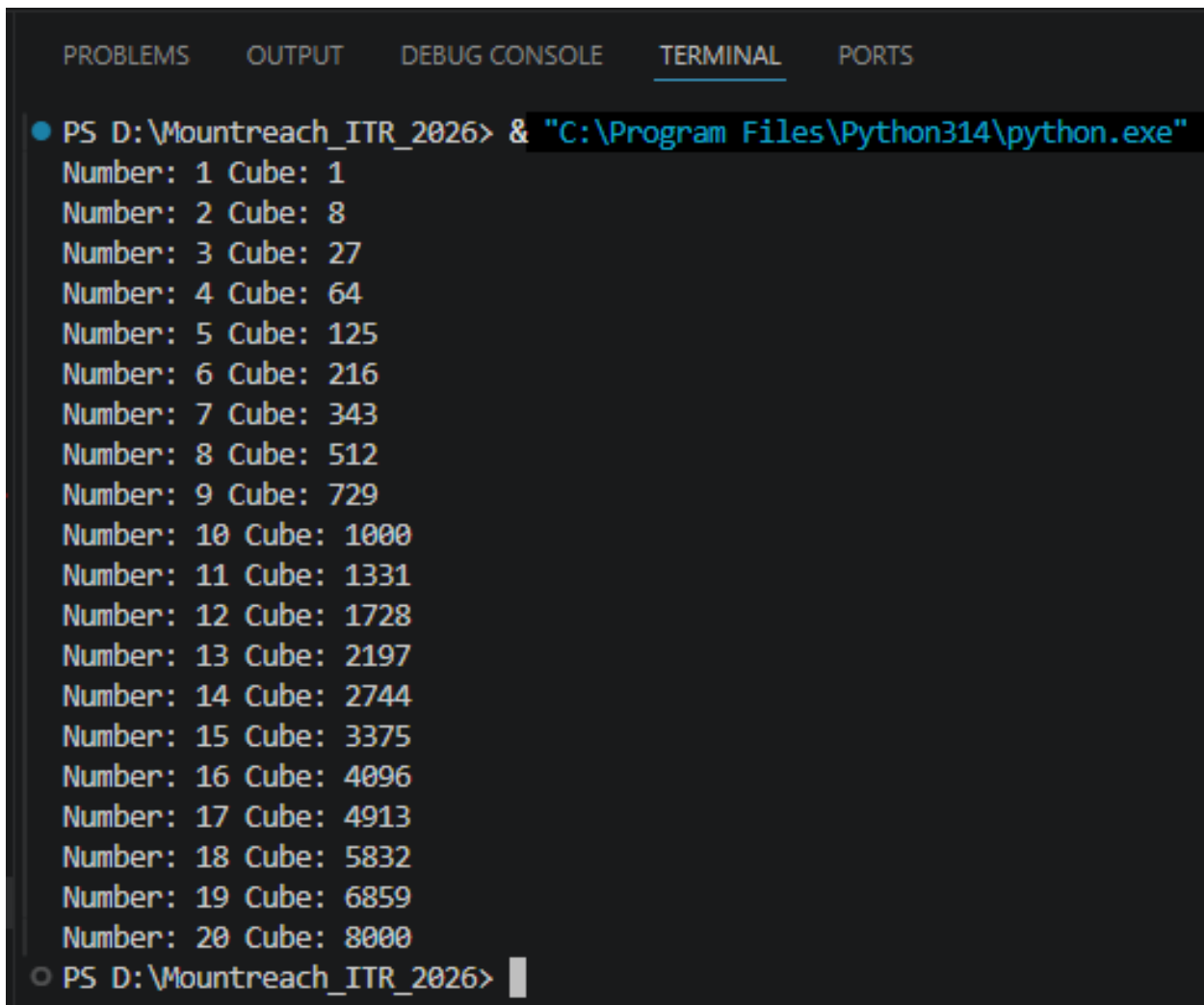
5. While Loop

Write a program using while loop to print the following:

- Numbers from 1 to 20.
- Also print the cube of each number (i.e., 1^3 , 2^3 , 3^3 , ...).

Stop when the number exceeds 20.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Number: 1 Cube: 1
Number: 2 Cube: 8
Number: 3 Cube: 27
Number: 4 Cube: 64
Number: 5 Cube: 125
Number: 6 Cube: 216
Number: 7 Cube: 343
Number: 8 Cube: 512
Number: 9 Cube: 729
Number: 10 Cube: 1000
Number: 11 Cube: 1331
Number: 12 Cube: 1728
Number: 13 Cube: 2197
Number: 14 Cube: 2744
Number: 15 Cube: 3375
Number: 16 Cube: 4096
Number: 17 Cube: 4913
Number: 18 Cube: 5832
Number: 19 Cube: 6859
Number: 20 Cube: 8000
○ PS D:\Mountreach_ITR_2026> |
```

Program Code:

```
# Initialize The Counter Variable To 1
number = 1

# The While Loop Runs As Long As Number Is 20 Or Less
while number <= 20:

    # Cube Of The Current Number
    cube = number ** 3

    # Print The Number And Its Cube Side By Side
    print("Number:", number, "Cube:", cube)

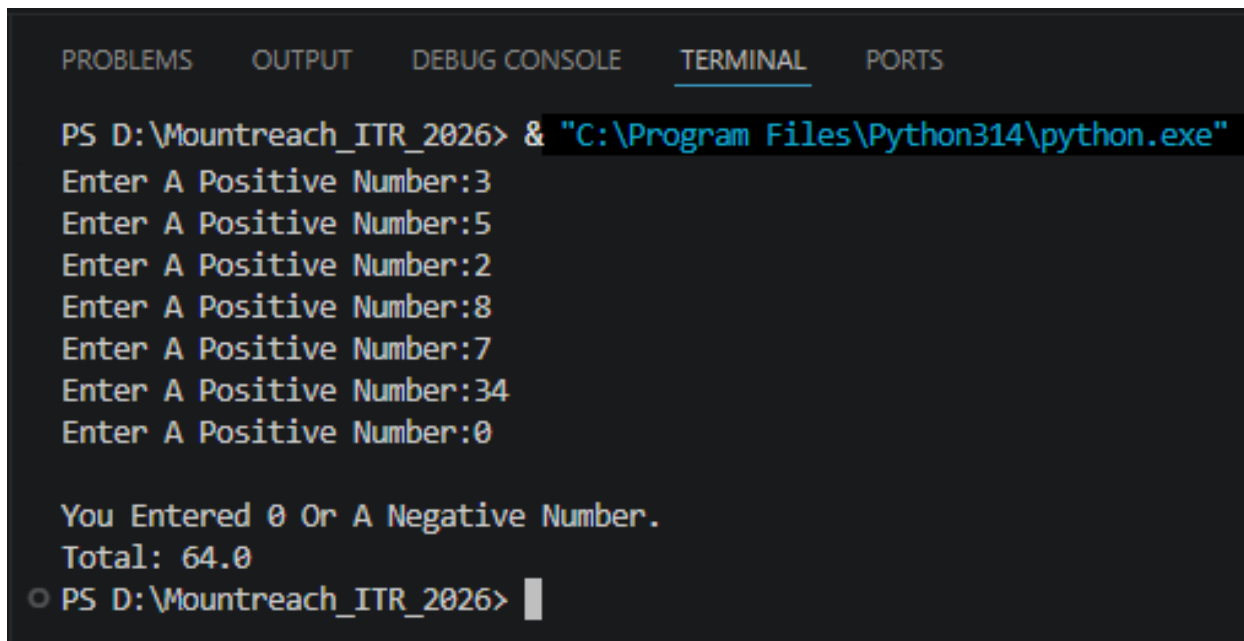
    # Increment The Counter By 1 To Move To The Next Number
    number = number + 1
```

6. while Loop - User Controlled

Write a program that repeatedly asks the user to enter a positive number.

Keep adding the entered numbers to a running total. Stop when the user enters a negative number or zero. Finally, print the total sum.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A Positive Number:3
Enter A Positive Number:5
Enter A Positive Number:2
Enter A Positive Number:8
Enter A Positive Number:7
Enter A Positive Number:34
Enter A Positive Number:0

You Entered 0 Or A Negative Number.
Total: 64.0
PS D:\Mountreach_ITR_2026> 
```

Program Code:

```
# Initialize The Total To 0
total = 0

# Take Input From The User For The First Number Before Entering The Loop
number = float(input("Enter A Positive Number:"))

# Keep Looping Until The User Enters A Positive Number
while number > 0:

    # Add The Entered Number To The Total
    total = total + number

    # Ask The User For The Next Number
    number = float(input("Enter A Positive Number:"))

print("\nYou Entered 0 Or A Negative Number.")
print("Total:", total)
```

7. Nested if Statement

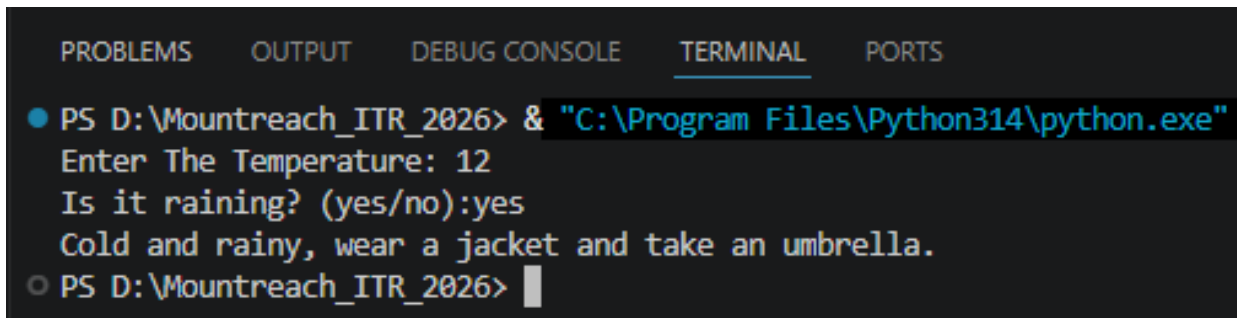
Write a program that takes two inputs:

- temperature (integer)
- is_raining (enter "yes" or "no")

Then give advice using nested if:

- If temperature > 30 and not raining → "Hot day, wear light clothes."
- If temperature > 30 and raining → "Hot and rainy, carry umbrella."
- If temperature < 15 and raining → "Cold and rainy, wear jacket and take umbrella."
- Else give a suitable message.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter The Temperature: 12
Is it raining? (yes/no):yes
Cold and rainy, wear a jacket and take an umbrella.
○ PS D:\Mountreach_ITR_2026> 
```

Program Code:

```
# Take Inputs From The User
temperature = int(input("Enter The Temperature: "))
is_raining = input("Is it raining? (yes/no):")

# Outer if: Check If Temperature Is Above 30 (Hot)
if temperature > 30:

    # Inner if: Check Rain Condition When It Is Hot
    if is_raining == "no":
        # If Hot & No Rain
        print("Hot day, wear light clothes.")
    else:
        # Or Else Hot & Raining
        print("Hot and rainy, carry an umbrella.")

# Outer elif: Check If Temperature Is Below 15 (Cold)
```



```
elif temperature < 15:

    # Inner if: Check Rain Condition When It Is Cold
    if is_raining == "yes":
        # If Cold & Raining
        print("Cold and rainy, wear a jacket and take an umbrella.")
    else:
        # Or Else Cold But No Rain
        print("Cold day, wear a warm jacket.")

# Outer else: Temperature Is Between 15 & 30
else:
    # Inner if: Check Rain Condition For Mild Weather
    if is_raining == "yes":
        # If Mild Temperature But Raining
        print("Mild but rainy, a light raincoat will do.")
    else:
        # Or Else Mild Temperature But No Rain
        print("Nice weather, enjoy your day!")
```

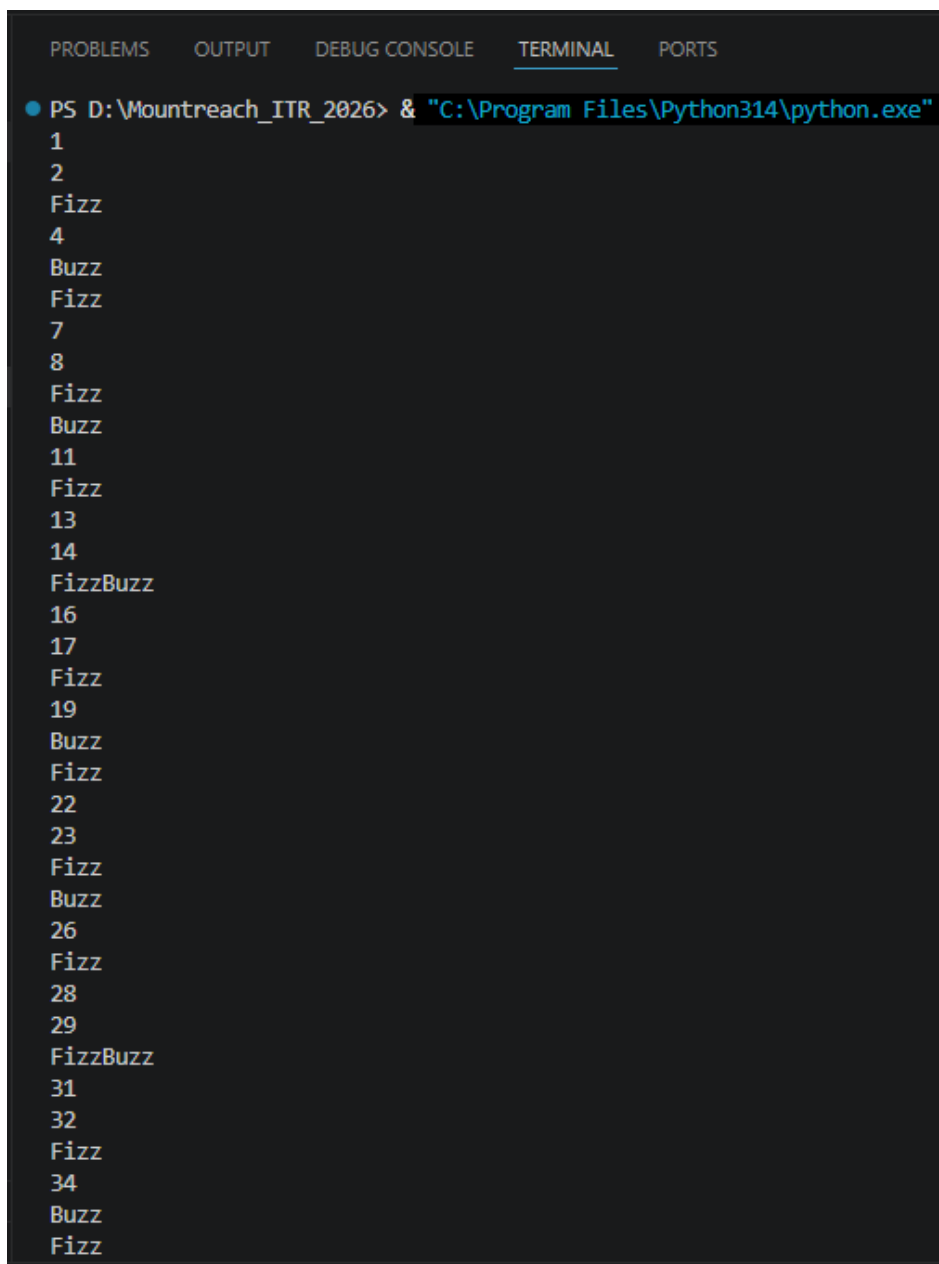
8. for Loop + if – elif - else

Write a program using for loop that goes from 1 to 40 and:

- Prints "Fizz" if number is divisible by 3
- Prints "Buzz" if number is divisible by 5
- Prints "FizzBuzz" if divisible by both 3 and 5

Otherwise prints the number

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
Fizz
22
23
Fizz
Buzz
26
Fizz
28
29
FizzBuzz
31
32
Fizz
34
Buzz
Fizz
```

Program Code:

```
# Starts At 1, Stops Before 41
for number in range(1, 41):

    # Check Divisibility By Both 3 & 5
    if number % 3 == 0 and number % 5 == 0:
        print("FizzBuzz")          # If Divisible By Both 3 & 5

    # Check Divisibility By 3 Only
    elif number % 3 == 0:
        print("Fizz")              # If Divisible By 3 Only

    # Check Divisibility By 5 Only
    elif number % 5 == 0:
        print("Buzz")              # If Divisible By 5 Only

    else:
        print(number)              # Or Else Just Print The Number
```

9. Menu Driven Program

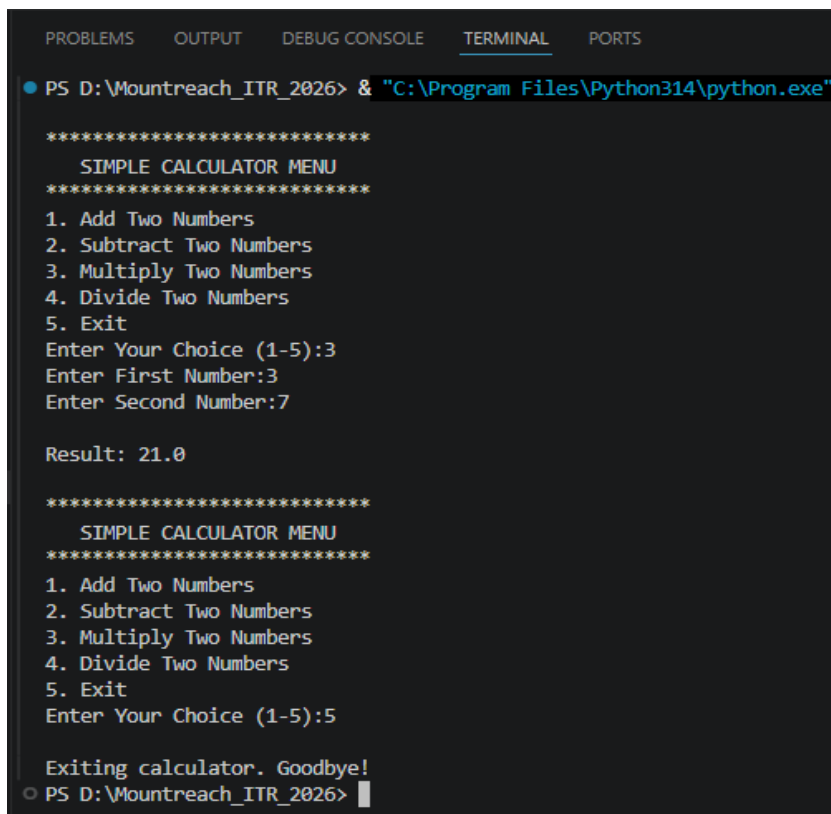
Create a simple calculator using while loop and if-elif-else:

Display this menu repeatedly:

```
1. Add two numbers
2. Subtract two numbers
3. Multiply two numbers
4. Divide two numbers
5. Exit
```

- Ask user for choice.
- Take two numbers as input.
- Perform the operation.
- Continue until user chooses 5 (Exit).
- Handle division by zero case.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
*****
SIMPLE CALCULATOR MENU
*****
1. Add Two Numbers
2. Subtract Two Numbers
3. Multiply Two Numbers
4. Divide Two Numbers
5. Exit
Enter Your Choice (1-5):3
Enter First Number:3
Enter Second Number:7

Result: 21.0

*****
SIMPLE CALCULATOR MENU
*****
1. Add Two Numbers
2. Subtract Two Numbers
3. Multiply Two Numbers
4. Divide Two Numbers
5. Exit
Enter Your Choice (1-5):5

Exiting calculator. Goodbye!
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
# The while Loop Keeps The Menu Running Until The User Chooses To Exit
while True:
```

```
    # Display The Menu Repeatedly
```

```
    print("\n*****")
    print("    SIMPLE CALCULATOR MENU    ")
    print("*****")
    print("1. Add Two Numbers")
    print("2. Subtract Two Numbers")
    print("3. Multiply Two Numbers")
    print("4. Divide Two Numbers")
    print("5. Exit")
```

```
    # Take Input From User For Menu Choice
```

```
    choice = int(input("Enter Your Choice (1-5):"))
```

```
    # Exit The Program If The User Chooses 5
```

```
    if choice == 5:
```

```
        print("\nExiting calculator. Goodbye!")
```

```
        break          # break Exits The While Loop Immediately
```

```
    # For Choices 1 To 4, Take Two Numbers As Input
```

```
    elif choice in [1, 2, 3, 4]:
```

```
        num1 = float(input("Enter First Number:"))
```

```
        num2 = float(input("Enter Second Number:"))
```

```
    # Perform The Operation Based On The User's Choice
```

```
    if choice == 1:
```

```
        result = num1 + num2          # Addition
```

```
        print("\nResult:", result)
```

```
    elif choice == 2:
```

```
        result = num1 - num2          # Subtraction
```

```
        print("\nResult:", result)
```

```
    elif choice == 3:
```

```
        result = num1 * num2          # Multiplication
```

```
        print("\nResult:", result)
```

```
    elif choice == 4:
```

```
        # Handle Division By Zero Before Performing The Operation
```

```
        if num2 == 0:
```

```
            print("\nError: Cannot Divide By Zero!")
```

```
        else:
```

```
            result = num1 / num2      # Division
```

```
            print("\nResult:", result)
```

```
# Handle Any Invalid Menu Choice (Not 1 To 5)
else:
    print("\nInvalid Choice! Please Enter A Number Between 1 And 5.")
```

10. Debugging

The following code contains several errors. Correct the program and add meaningful comments explaining each fix:

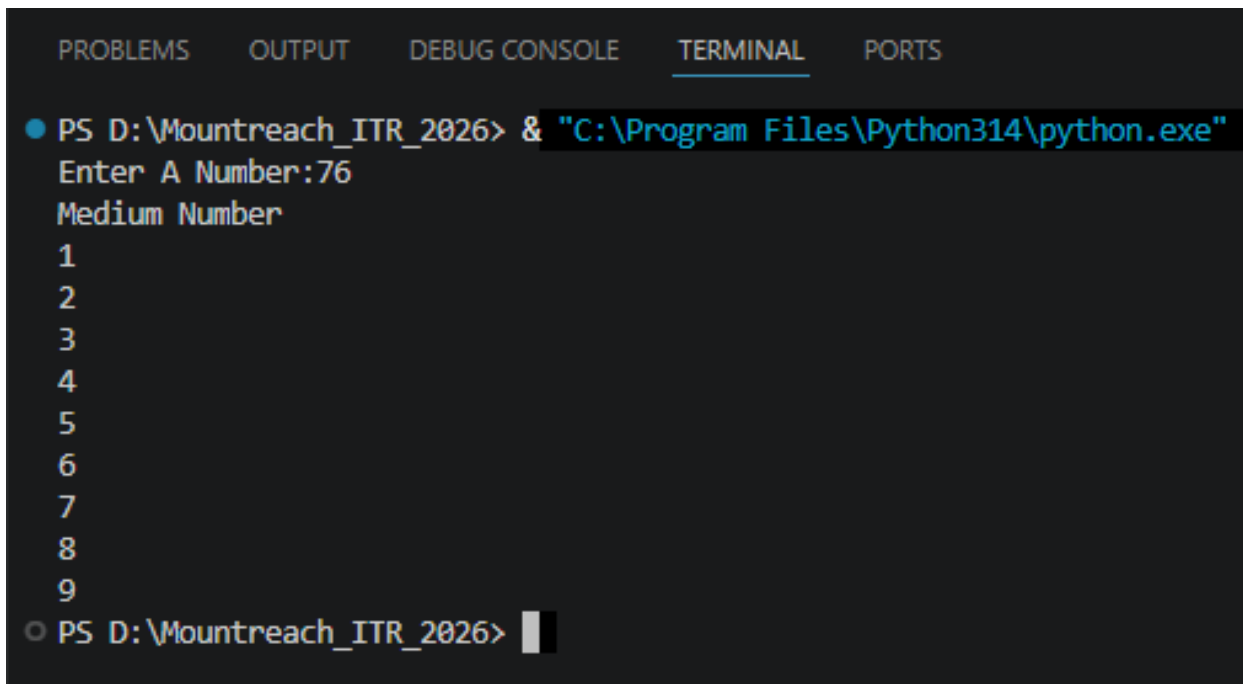
```
num = input("Enter a number: ")

if num > 100
    print("Large number")
elif num > 50

    print("Medium number")
else

count = 1
while count < 10
print(count)
    +
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A Number:76
Medium Number
1
2
3
4
5
6
7
8
9
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
num = int(input("Enter A Number:")) # int() To Convert String Input To integer

if num > 100:                        # Added missing colon (:)
    print("Large Number")

elif num > 50:                      # Added missing colon (:)
    print("Medium Number")

else:                               # Added missing colon (:)
    print("Small Number")

count = 1
while count < 10:                  # Added missing colon (:)
    print(count)
    count = count + 1              # Fixed count+1 To count=count+1 To
Increment The Variable
```